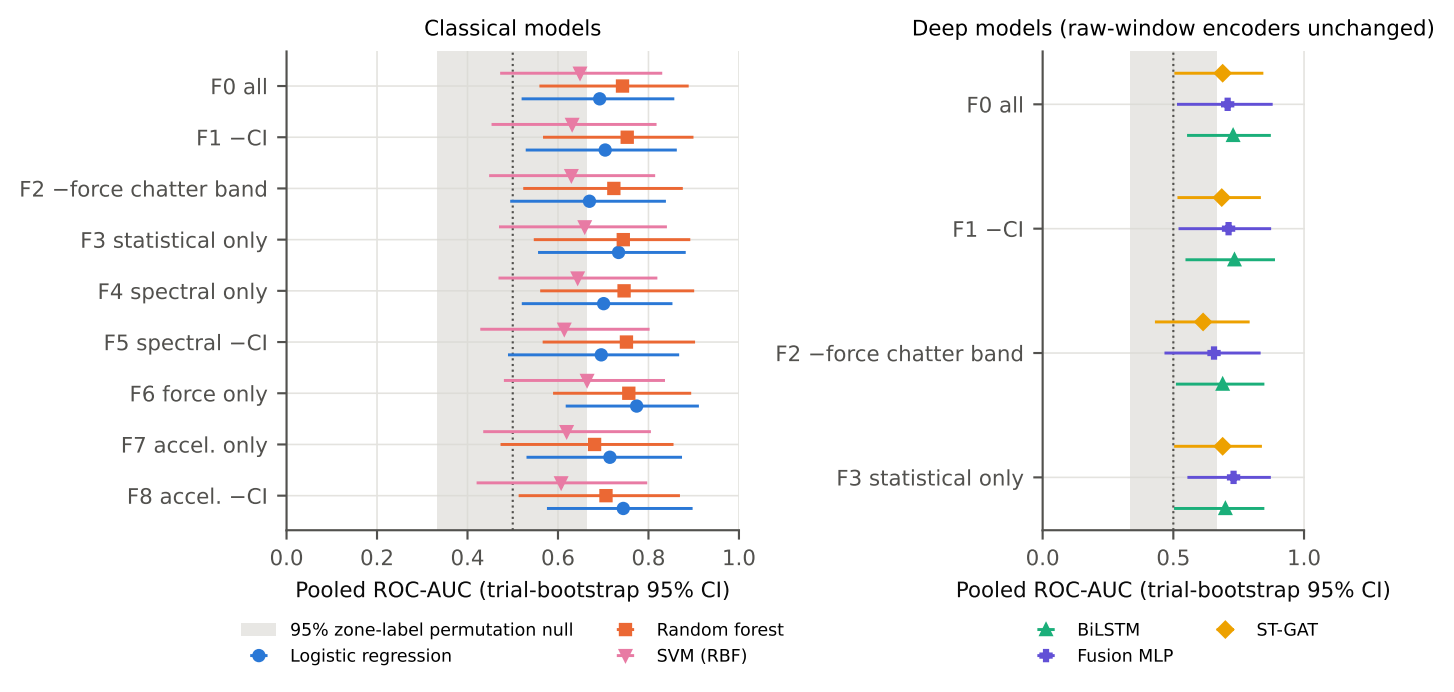
Classical models
F0 all
F1 −CI
F2 −force chatter band
F3 statistical only
F4 spectral only
F5 spectral −CI
F6 force only
F7 accel. only
F8 accel. −CI
0.0
0.2
0.4
0.6
0.8
1.0
Pooled ROC-AUC (trial-bootstrap 95% CI)
95% zone-label permutation null
Logistic regression
Random forest
SVM (RBF)
Deep models (raw-window encoders unchanged)
F0 all
F1 −CI
F2 −force chatter band
F3 statistical only
0.0
0.5
1.0
Pooled ROC-AUC (trial-bootstrap 95% CI)
BiLSTM
Fusion MLP
ST-GAT